MAIS PAC

IPhO 02.16: Heat Transfer, Phase Transitions and Radiation · Worked Solutions

Heat Transfer, Phase Transitions and Radiation: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

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Solution 1 — Heating water. Return →

(a) Q = mcΔ T = (1.5)(4200)(80) = 5.0×10^5 {J}.

(b) t = Q/P = 5.04×10^5/2000 = 2.5×10² {s} — about four minutes.

(c) The kettle itself heats up, some heat is lost to the room, the element is not perfectly efficient, and the water evaporates a little. In addition, the rated power is delivered to the element, not to the water, so the transfer efficiency is below 100%. **Check:** 5.0×10^5 {J} is about 0.14 {kWh}, and a 2 {kW} kettle delivers that in 4 minutes.

Solution 2 — Melting ice. Return →

(a) Energy to melt the ice: Q_1 = (0.50)(3.34×10^5) = 1.67×10^5 {J}. Energy released by the water cooling to 0°{C}: Q_2 = (2.0)(4200)(25) = 2.1×10^5 {J}. Since Q_2 > Q_1, all the ice melts, and the surplus (2.1 - 1.67)×10^5 = 4.3×10^4 {J} warms 2.5 {kg} of water:

$$\Delta T = (4.3\times10^4)/((2.5)(4200)) = 4.1\ \{K\}, \tag{1}$$

so the final temperature is about 4.1°{C}.

(b) Confirmed above: 2.1×10^5 > 1.67×10^5 {J}.

(c) With 0.30 {kg} the melting energy is 1.0×10^5 {J}, still less than 2.1×10^5, so the final temperature is Δ T = (2.1-1.0)×10^5/(2.3×4200) = 11 {K} above zero, i.e. about 11°{C}. **Check:** less ice gives a warmer mixture, as expected; the check on the assumption is always whether the available energy exceeds the latent heat.

Solution 3 — Conduction through a window. Return →

(a) A = 1.2×1.5 = 1.8 {m²}, so

$$(dQ)/(dt) = (kA\Delta T)/(d) = ((0.80)(1.8)(20))/(0.0040) = 7.2\times10^3\ \{W\} = 7.2\ \{kW\}. \tag{2}$$

(b) In a day, E = 7200×86,400 = 6.2×10^8 {J} = 173 {kWh}.

(c) Setting d/k equal for air and glass: d_({air}) = 0.026×(0.0040/0.80) = 1.3×10^(-4) {m} — a tenth of a millimetre of still air is as good as 4 {mm} of glass. That is why double glazing uses a thick gap; in practice the air gap must be wide enough that convection does not short-circuit it. **Check:** the window loses about 7 {kW}, which would be a large fraction of a domestic heating system — the reason glazing matters.

Solution 4 — Radiation from a person. Return →

(a) T = 306 {K}:

$$P = \varepsilon\sigma\, AT^4 = (0.97)(5.67\times10^{-8})(1.8)(306)^4 = 8.8\times10^{2}\ \{W\}. \tag{3}$$

(b) The room at 293 {K} returns

$$P_{\{in\}} = (0.97)(5.67\times10^{-8})(1.8)(293)^4 = 7.4\times10^{2}\ \{W\}, \tag{4}$$

so the net loss is about 1.4×10^{2} {W}.

(c) The body generates roughly 100 {W} at rest, and evaporative loss and convection make up the rest. Cold walls lower T_0 in the fourth-power term, so the net radiative loss grows even if the air is warm — which is why a room with single glazing feels cold at the same air temperature. **Check:** the ratio $P/P_{\{in\}} = (306/293)^4 = 1.19$, and $0.19\times880 = 170$ {W}, consistent with the difference calculated directly.

Solution 5 — The Sun and the Earth. Return →

(a) $A = 4\pi R^2 = 4\pi(6.96\times10^{8})^2 = 6.09\times10^{18}\ \{m^2\}$, so

$$P = \sigma\, AT^4 = (5.67\times10^{-8})(6.09\times10^{18})(5800)^4 = 3.9\times10^{26}\ \{W\}. \tag{5}$$

(b) At $r = 1.5\times10^{11}$ {m}, the intensity is $S = P/(4\pi r^2) = 3.9\times10^{26}/(2.83\times10^{23}) = 1.4\times10^{3}\ \{W,m^{-2}\}$.

(c) Energy balance for a black sphere absorbing over its cross-section πR^2 and radiating over its whole surface $4\pi R^2$:

$$\pi R^2 S = 4\pi R^2 \sigma T^4 \quad T = \left(\frac{S}{4\sigma}\right)^{1/4} = \left(\frac{1360}{4(5.67\times10^{-8})}\right)^{1/4} = 279\ \{K\} = 6^{\circ}\{C\}. \tag{6}$$

Adding the albedo 0.30 lowers it to about 255 {K}; the real mean is 288 {K} because the atmosphere absorbs and re-radiates infrared, the greenhouse effect. **Check:** the calculation reproduces the measured solar constant to about 3%, and the 33 {K} gap is the standard greenhouse figure.

Solution 6 — Newton's law of cooling. Return →

(a) Using excess temperatures: 90-20 = 70 and 70-20 = 50, so 50 = 70e^(-300k) and k = (ln (70/50))/(300) = 1.12×10^(-3) {s^(-1)}.

(b) After 15 {min} total: excess = 70e^(-1.12×10^(-3)×900) = 70(0.365) = 26 {K}, so T = 46°{C}.

(c) 30°{C} means an excess of 10 {K}: t = (ln 7)/(1.12×10^(-3)) = 1.7×10³ {s} ≈ 29 {min}. The model fails at large excess because radiation (T^4) takes over and because evaporation is significant for hot water. **Check:** the model predicts an asymptote at 20°{C}, never below, which is right but approached more slowly than reality allows.

Solution 7 — Two layers of insulation. Return →

(a) Resistances per unit area: brick 0.10/0.84 = 0.119, insulation 0.10/0.040 = 2.50 {m²,K,W^(-1)}; total 2.62. Then

$$\{\text{flux}\} = (20)/(2.62) = 7.6\ \{\text{W,m}^{(-2)}\}. \tag{7}$$

(b) The drop across the brick is 7.6×0.119 = 0.91 {K}, so the interface is at 19.1°{C}.

(c) The insulation has twenty-one times the resistance of the brick, so it takes twenty-one times the temperature drop: 19.1 {K} across the insulation and 0.9 {K} across the brick. **Check:** summing the drops, 0.91 + 19.1 = 20 {K}, as required.

Solution 8 — A calorimetry puzzle. Return →

(a) Warming the ice: (0.20)(2100)(10) = 4.2×10³ {J}. Melting it: (0.20)(3.34×10^5) = 6.7×10^4 {J}. Total 7.1×10^4 {J}.

(b) The water cooling to 0°{C} releases (1.0)(4200)(30) = 1.26×10^5 {J}.

(c) Since 1.26×10^5 > 7.1×10^4, all the ice melts with 5.5×10^4 {J} to spare, which warms 1.2 {kg} of water by 5.5×10^4/(1.2×4200) = 11 {K}. The final state is 1.2 {kg} of liquid water at 11°{C}. **Check:** the surplus is about half of the cooling energy, and 11°{C} is about a third of 30°{C} — consistent with the water being diluted by the meltwater.

Solution 9 — The greenhouse effect. Return →

(a) Absorbed power = S(1-α)π R², radiated power = 4π R²σ T^4; equating them gives T^4 = S(1-α)/(4σ).

(b) T = ((1360(0.70))/(4(5.67×10^(-8))))^(1/4) = (4.20×10^9)^(1/4) = 255 {K} = -18°{C}.

(c) The observed mean surface temperature is about 288 {K}. A single-layer atmosphere that is transparent to visible light but absorbs all infrared re-radiates half downwards, so the surface must radiate twice as much: T^4 doubles and T rises by a factor 2^(1/4) = 1.19, giving 303 {K} = 30°{C} — too warm, which shows the single-layer model is crude but captures the mechanism. **Check:** the 33 {K} discrepancy is exactly the greenhouse effect, and doubling T^4 predicts 45 {K}, of the right order.

Solution 10 — Design of a hot-water tank. Return →

(a) Side = 2π rH = 2π(0.35)(1.5) = 3.30 {m²}; ends = 2π r² = 0.77 {m²}; total A = 4.07 {m²}.

(b) Using the mean radius of the insulation, about 0.39 {m}, A≈4.5 {m²}, and

$$(dQ)/(dt) = (kA\Delta T)/(d) = ((0.035)(4.5)(40))/(0.080) = 79\ \{W\}. \tag{8}$$

(c) In a day the loss is 79×86,400 = 6.8×10^6 {J}, and cooling 200 {kg} of water by that much gives Δ T = 6.8×10^6/((200)(4200)) = 8.1 {K} — about 8°{C} per day. Real tanks lose more because of the metal walls (which conduct), pipe connections (thermal bridges) and convection in the air gap. **Check:** 79 {W} is a plausible standing loss for a domestic tank, and 8°{C} per day matches the "loses a few degrees a day" rule of thumb.